

Abena Boadi-Agyemang

CONTACT INFORMATION

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RESEARCH STATEMENT

I am a roboticist and researcher specializing in emerging physical and digital technologies to support the wellbeing of diverse people. My research focuses on creating assistive technologies to support adults with disabilities and chronic health conditions with their daily tasks. I have worked on intelligent and autonomous systems across safety-critical domains, such as autonomous mobility and health tech. My competencies include mixed-methods human studies, full-stack systems engineering, and machine learning. In the past, my research areas included embedded systems and mechanical design for inclusive haptic interfaces, novel wearable devices, and spatial computing technologies.

EDUCATION

Carnegie Mellon University , Doctor of Philosophy in Robotics, School of Computer Science <i>Advisor:</i> Aaron Steinfeld. <i>Depth:</i> Human-Robot Interaction; Sociotechnical Systems; Inclusive Design.	2026
Carnegie Mellon University , Master of Science in Robotics, School of Computer Science <i>Depth:</i> Human-Robot Interaction; Mechatronics; Haptics Interfaces. <i>Relevant Coursework:</i> Robot Learning; Kinematics, Dynamics, and Control; Sensing and Sensors.	2025
Stanford University , Bachelor of Science in Mechanical Engineering, School of Engineering <i>Depth:</i> Dynamics Systems and Control.	2021

RESEARCH POSITIONS

TBD Lab , Robotics Institute, Carnegie Mellon University <i>Graduate Researcher</i> , <i>Advisor:</i> Aaron Steinfeld - Pioneered participatory approaches to involve non-expert adults with disabilities in cooperative design (“co-design”) processes to ensure service robots can provide equitable support; led multi-scale human evaluations of service robotics in 3 safety-critical domains relevant to disabled stakeholders: personal health, assistive navigation, and last-mile delivery. - Facilitated 40+ design workshops with disabled stakeholders focused on varied-fidelity participatory prototyping across the robotics stack to generate community-informed technical recommendations and highlight sociotechnical risks. - Engineered impedance-type haptic devices to render virtual surfaces for non-visual communication of STEM concepts to blind and low vision (BLV) K-12 students. - Conducted in-depth qualitative studies with K-12 educators (teachers of visually impaired students) to understand gaps and barriers in providing non-visual aids to BLV students.	2021 – 2026
SHAPE Lab , Mechanical Engineering Department, Stanford University <i>Undergraduate Researcher</i> , <i>Advisors:</i> Dr. Alexa F. Siu; Dr. Elyse D. Z. Chase; <i>PI:</i> Dr. Sean Follmer Researched and architected force- and tactile-feedback devices to enable users who are visually impaired to interact with graphical information and virtual objects.	2019 – 2021
Clayman Institute for Gender Research , Stanford University	2018 – 2019

Undergraduate Research Assistant, Advisor: Dr. Melissa V. Abad

- Conducted content analyses of the gendered-language used in Fortune 500 company leadership statements to compare the downstream effects on workplace diversity; produced frequency database to guide organizational inclusivity interventions.

Graduate School of Education, Stanford University

2017 – 2018

Undergraduate Research Assistant, Advisor: Dr. John Willinsky, Public Knowledge Project (PKP) Founder

- Executed forensic website audits of predatory journals (from “Beall’s List”) utilizing Open Journal Systems (OJS), an open-source publishing software developed by the PKP; prepared a contact database to elicit journals’ participation in integrity interventions.

OTHER PROFESSIONAL EXPERIENCE

Robots for All Initiative

Mar 2026 – Present

Co-founder, Robot Accessibility Expert, & Core Leadership Team Member

- Generate and advise on accessibility and usability protocols to define best practices for safe, inclusive physical human-robot interactions; write grants to acquire funding to co-create guidelines with stakeholder groups.
- Co-lead the inaugural Robots for All Consortium, facilitating cross-organizational collaboration between industry leaders, academic researchers, and disabled stakeholders to inform national robot accessibility guidelines.

Robotics Institute, Carnegie Mellon University

Jul 2023 – May 2025

Course Assistant & Instructor. Courses: Sensing & Sensors (Grad-Level), Human-Robot Interaction (UGrad-Level)

- Directed & advised research for 20+ cross-functional teams (Robotics, Engineering, Computer Science) across 19 distinct projects, providing oversight on methodology, experimental design, & data analysis.
- Developed & delivered advanced lecture on sensor fusion & state estimation (Odometry & Kalman Filtering) to 20+ graduate students, addressing gaps in technical understanding.

Autodesk

Jul 2020 – Dec 2020

Fusion Catalyst Intern

- Drove on-campus adoption of Fusion 360 (Cloud-based CAD/CAM & novel generative AI design integration) by providing expert technical training to student engineers.
- Optimized resource allocation by aligning Autodesk technical experts with student engineering organizations, ensuring efficient software utilization.

AWARDS & HONORS

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- **Best Paper Honorable Mention Award** 2026
Top 3% at ACM/IEEE HRI Design Track.
 - **HRI Pioneer** 2026
Recognized as 1 of 22 elite global HRI researchers for contributions to the future of Human-Robot Interaction.
 - **Cadence Future Innovators Scholarship** 2023
Award (\$5k) to a student who exemplifies leadership and demonstrates a passion for technology.
 - **Carnegie Mellon Uber Presidential PhD Fellow** 2022 – 2023
 - Honorable Mention **Hoefer Prize for Writing in the Major**
Stanford Mechanical Engineering Department’s Nominee for outstanding undergraduate Writing in the Major (WIM) capstone.
 - **University of Washington College of Engineering Dean’s Fellowship (declined)** 2021
Graduate fellowship award for top incoming doctoral applicants (\$10k/yr) (Nominated by HCDE).

- **University of Washington Top Scholar Fellowship** (*declined*) 2021
Award to attract outstanding students to the University of Washington (\$2.5k)
- **Stanford Mechanical Engineering Summer Undergraduate Research Institute Fellowship** 2019, 2020
Research fellowship (\$7.5k) to conduct and present a ten-week, original research project.
- **Academic Award of Excellence** 2018, 2019
Recipient for outstanding academic achievements, awarded by Stanford University Black Community Services Center.
- **Botswana Top Achiever's Award and Scholarship** 2017, 2021
Highly selective national merit award granted by the Ministry of Education, fully funding undergraduate (Stanford) and graduate (CMU) studies for top-ranked national scholars.
- **American University Preparatory School Global Leadership Merit Scholarship** 2014-2017
Three-year academic scholarship for a student with exemplary leadership qualities.

PUBLICATIONS [*denotes equal contribution. [Google Scholar](#) for full papers.]

Journal Articles and Book Chapters (Peer Reviewed)

1. Shih, K. *, **Boadi-Agyemang, A. ***, Carter, E. J., Steinfeld, A. (2025). A Multi-Method Investigation of Guide Robot Characteristics for Blind and Low Vision Users. In *ACM Transactions on Human-Robot Interaction*.
2. Siu, A. F., Chase, E. D. Z., Kim, G. S. H., **Boadi-Agyemang, A.**, Gonzalez, E. J., and Follmer, S. (2021). Haptic Guidance to Support Design Education and Collaboration for Blind and Visually Impaired People. In *Plattner H., Design Thinking Research. Understanding Innovation. Springer*.

Conference and Short Papers (Peer Reviewed)

3. **Boadi-Agyemang, A.**, Adjagbodjou, A. D., Cunningham, J. L. (2026). Relational and Somatic Engagements with Anxious and Depressed Black Women: A Speculative Design Approach to Social Robot-Mediated Mental Well-being. In *Proceedings of the 2026 Conference on Creativity and Cognition*.
4. **Boadi-Agyemang, A.** (2026). Robot Design at the Margins: The Lived Experience as a Participatory Approach to Co-designing with Disabled Adults. In *Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*.
5. **Boadi-Agyemang, A.**, Moharana, S., Bennett, C. L., Carter, E. J., Carrington, P., & Steinfeld, A. (2026). Pluriversal Approach to Co-designing Delivery Robots with People with Disabilities. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. **[Best Paper Honorable Mention Award, Top 3%]**.
6. **Boadi-Agyemang, A.**, Schaldenbrand, P., Misra, V., Carter, E. J., Oh, J., & Steinfeld, A. (2025). If I Move, Do You Move? Investigating the Role of Interpersonal Synchrony in Human-Robot Joint Painting. In *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*.
7. **Boadi-Agyemang, A. *** and Park, M. *. (2024). Designing Interactive Agents to Support Emotion Regulation in the Workplace through Guided Art-Making. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*.
8. Han, V. Y., **Boadi-Agyemang, A.**, Lin, Y., Lindlbauer, D., and Ion, A. (2023). Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices. In *The 35th Annual ACM Symposium on User Interface Software and Technology (UIST)*.
9. **Boadi-Agyemang, A.**, Carter, E. J., Siu, A. F., Steinfeld, A., and Martinez, M. O. (2023). Understanding Experiences, Attitudes, and Perspectives towards Designing Interactive Creative Tools for Teachers of Visually Impaired Students. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.

10. Kim, L. H.*, **Boadi-Agyemang, A.***, Siu, A. F., and Tang, J. (2020). When to Add Human Narration to Photo-Sharing Social Media. In *The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.
11. Chase, E. D. Z., Siu, A. F., **Boadi-Agyemang, A.**, Kim, G. S. H., Gonzalez, E. J., and Follmer, S. (2020). PantoGuide: A Haptic and Audio Guidance System to Support Tactile Graphics Exploration. In *The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.

Workshops and Technical Reports (Non-Archival, Lightly Reviewed)

12. Carter, E. J., **Boadi-Agyemang, A.**, Moharana, S., Carrington, P., & Steinfeld, A. (2026). Rendezvous with Automation at the Curb. *UCLA: Mobility Center of Excellence*.
13. **Boadi-Agyemang, A.***, Moharana, S.* (2025). The Case for Em(body)ed Approaches to Disability-Centered Interaction Design. "Body Politics: Unpacking Tensions and Future Perspectives for Body-Centric Design Research in HCI" at *ACM Conference on Human Factors (CHI)*.
14. **Boadi-Agyemang, A.***, Shih, K.* and Steinfeld, A. (2023). Discovering User Needs and Preferences for Guide Robots: Challenges and Preliminary Insights. In Workshop "Assistive Robots for Citizens" at *IEEE/RSJ Conference on Intelligent Robots and Systems (IROS)*.

Theses

15. **Boadi-Agyemang, A.** (2026). From Margin to Center: Designing Inclusive and Equitable Service Robots with Disabled Adults. *Doctoral Thesis. Carnegie Mellon University*.
16. **Boadi-Agyemang, A.** (2024). Simulated Encounters of the Third Kind: Scenario-Based Approach to Designing Guide Robots. *Master's Thesis. Carnegie Mellon University*.
17. Perez, G. A.*, **Boadi-Agyemang, A.***, Mhatre, K.*, Wang, J.* (2021). Using a Vibrotactile Stimulation Device to Improve Knee Extension During Stance Phase in Children with Spastic Cerebral Palsy. *Undergraduate Capstone, Stanford University*. [Nominated for Hoefer Prize for Writing in the Major].

RESEARCH SKILLS

Human-Centered Research Methods & Analysis

- **Usability Evaluations:** Cognitive Interviews (Think-Alouds), Contextual Inquiry, A/B Testing, Heuristic Evaluation.
- **Qualitative Methods:** In-depth Interviews, Diary Studies, Focus Groups.
- **Survey & Sampling Methodology:** Large-Scale Online Surveys (Qualtrics, Prolific), Non-Probability Sampling (Purposive, Convenience, Snowball).
- **Participatory Design (PD):** Community-Based PD Research, Co-design Workshops, Capacity Building/Mutual Learning.
- **Qualitative Analysis:** Reflexive Thematic Analysis, Affinity Diagramming, Content and Discourse Analysis.
- **Experimental Design & Statistics:** Factorial Study Design (Within/Between-Subjects, Latin-Square), Advanced Statistical Testing (REML Modeling, ANOVA, T-Tests).
- **Quant/Qual Data Analysis Software:** R, JMP, MATLAB, ATLAS.ti.
- **Databases:** MongoDB, SQLite.

TECHNICAL SYSTEM SKILLS

Robotics, Interactive AI, & Social Intelligence

- **Programming Languages:** Python, C, C++, Java.
- **Robotics, ML & Simulation Frameworks:** ROS/ROS2, PyTorch, TensorFlow, Gymnasium, MuJoCo, Simulink.
- **OS:** Linux (Ubuntu).
- **Robot & Machine Learning:** Reinforcement Learning, Imitation Learning, Supervised/Preference Learning (Bayesian Contextual Bandits, RLHF).
- **Perception & State Estimation:** OpenCV, Sensor Fusion, Kalman Filtering, Odometry.

- **Social Reasoning & Behavior:** Theory of Mind (ToM), Intent Modeling, Interpersonal Coordination (Synchrony, Behavioral Matching), User Behavior & Trust Modeling, Personalization.
- **Agentic AI & Speech Integration:** OpenAI API, ElevenLabs, gTTS.

Mechatronics, Web & Physical-Digital Prototyping

- **Embedded Hardware & Microcontrollers:** STM32, Arduino, Circuit Design, Embedded C/C++, Monkey C (Garmin).
- **Mechanical Design & Fabrication:** CAD/CAM (SolidWorks, Fusion 360), 3D Printing, Laser Cutting.
- **Web & Spatial Graphics:** Netlify, JavaScript (React.js, Node.js), Unity 3D, Processing.
- **UX & Design Prototyping:** Wireframing (Figma, Adobe XD), Journey Mapping (Miro).